Branch 2 (Degenerated 4-Bar) Analysis Report with Explicit Matrix Operations

Automated Generation System

March 13, 2026

1 Mathematical Formulation

For this specific branch, the topological constraint locks joints J_2, J_4, and J_6 to zero. The driving joint is J_3. The full kinematic constraint $T_1 \dots T_7 = I$ is reduced to three active variables (q_1, q_5, q_7). The velocity constraint yields $J_p \dot{\mathbf{q}}_p = -J_a \dot{q}_3$, where $J_p \in R^{6\times 6}$ is the passive Jacobian matrix obtained by removing the column corresponding to the driving joint J_3.

2 Example Matrix Calculation at $q_3 = 0^\circ$

The solved full joint configuration (in radians), showing the locked joints explicitly as 0, is:

$$\mathbf{q} = \begin{bmatrix} -0.0000 \\ 0.0000 \\ 0.0000 \\ 0.0000 \\ -0.0000 \\ 0.0000 \\ 0.0000 \end{bmatrix} \tag{1}$$

Based on this configuration, the full 6×7 Jacobian matrix J is evaluated as:

$$J = \begin{bmatrix} -0.000 & -0.000 & -0.000 & 2.500 & -0.000 & -2.500 & 0.000 \\ 0.000 & -1.000 & -0.000 & 3.000 & -0.000 & 2.000 & 0.000 \\ 0.000 & 0.000 & -3.000 & -0.000 & -0.500 & 0.000 & -2.000 \\ 0.000 & 0.000 & -0.000 & -0.000 & -0.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & -1.000 & -0.000 & -1.000 & 0.000 & 1.000 \\ 1.000 & 1.000 & 0.000 & -1.000 & 0.000 & 1.000 & 0.000 \end{bmatrix} \tag{2}$$

Extracting the columns of the passive joints yields $J_p \in R^{6\times 6}$:

$$J_p = \begin{bmatrix} -0.000 & -0.000 & 2.500 & -0.000 & -2.500 & 0.000 \\ 0.000 & -1.000 & 3.000 & -0.000 & 2.000 & 0.000 \\ 0.000 & 0.000 & -0.000 & -0.500 & 0.000 & -2.000 \\ 0.000 & 0.000 & -0.000 & -0.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & -0.000 & -1.000 & 0.000 & 1.000 \\ 1.000 & 1.000 & -1.000 & 0.000 & 1.000 & 0.000 \end{bmatrix} \tag{3}$$

Applying Singular Value Decomposition (SVD) on $J_p = U\Sigma V^T$:

$$U = \begin{bmatrix} -0.667 & 0.626 & -0.000 & -0.405 & 0.000 & 0.000 \\ -0.662 & -0.747 & 0.000 & -0.066 & -0.000 & -0.000 \\ 0.000 & 0.000 & 0.894 & 0.000 & -0.447 & -0.000 \\ -0.000 & -0.000 & -0.000 & -0.000 & -0.000 & -1.000 \\ -0.000 & -0.000 & -0.447 & -0.000 & -0.894 & 0.000 \\ 0.343 & -0.224 & 0.000 & -0.912 & 0.000 & 0.000 \end{bmatrix} \tag{4}$$

$$\Sigma = \begin{bmatrix} 4.190 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & 3.362 & 0.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & 2.236 & 0.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & 1.280 & 0.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & 1.118 & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 \end{bmatrix} \tag{5}$$

$$V^T = \begin{bmatrix} 0.082 & 0.240 & -0.953 & 0.000 & 0.164 & -0.000 \\ -0.067 & 0.156 & -0.134 & -0.000 & -0.976 & -0.000 \\ 0.000 & -0.000 & 0.000 & 0.000 & 0.000 & -1.000 \\ -0.713 & -0.661 & -0.232 & 0.000 & -0.025 & -0.000 \\ 0.000 & 0.000 & 0.000 & 1.000 & -0.000 & 0.000 \\ 0.693 & -0.693 & -0.139 & 0.000 & -0.139 & 0.000 \end{bmatrix} \tag{6}$$

3 Analysis Results & Curve Fitting

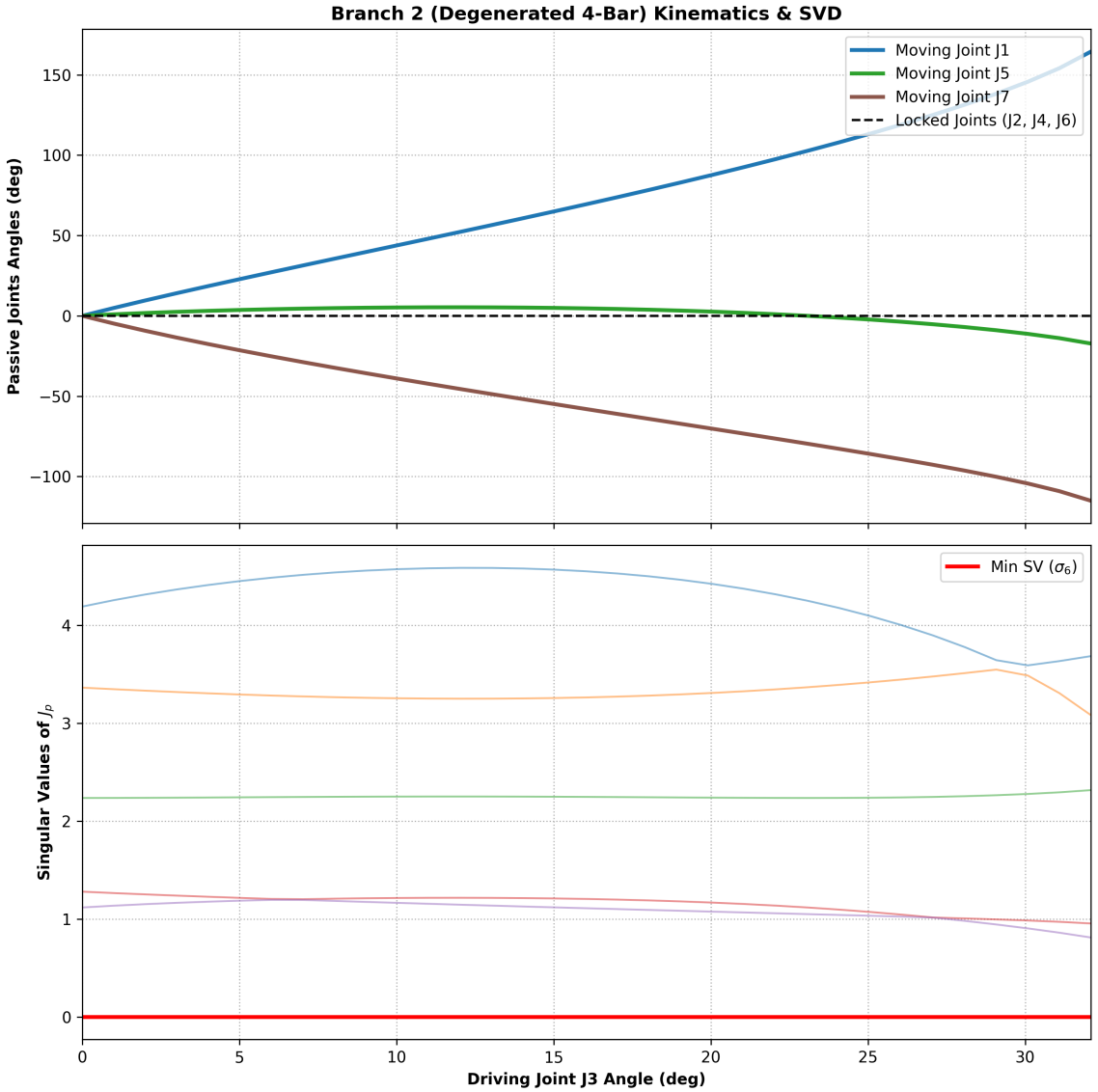


Figure 1: Kinematic mapping and singular value evolution driven by J_3 on Branch 2.

The Fourier series expressions for the **moving** passive joints are:

- **Joint J1:**
  $Q_{J1}(x) = +295246.095 - 433177.915\cos(x) - 97759.163\sin(x) + 161643.091\cos(2x) + 77370.455\sin(2x) - 23711.147\cos(3x) - 18904.486\sin(3x)$
- **Joint J5:**
  $Q_{J5}(x) = -115151.694 + 167713.559\cos(x) + 41668.084\sin(x) - 61192.262\cos(2x) - 32466.924\sin(2x) + 8630.367\cos(3x) + 7775.662\sin(3x)$
- **Joint J7:**
  $Q_{J7}(x) = -179423.214 + 264492.669\cos(x) + 55900.899\sin(x) - 100103.702\cos(2x) - 44706.010\sin(2x) + 15034.155\cos(3x) + 11079.610\sin(3x)$